EXPERIMENT NO.10

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Aim: Explore the Cryptocurrency Landscape

Theory:

This experiment explores the cryptocurrency landscape by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is **Chainlink (LINK)**, and data was collected from three reputable sources as per the lab requirement:

- **Financial Data (APIs)** – Services like **CoinGecko** and **CoinMarketCap** provided structured data on market capitalization, circulating supply, and historical 24-hour trading volumes, which are essential for calculating liquidity and asset dominance.
- **News & Social Platforms** – Platforms such as **CryptoPanic** (news aggregator) and **LunarCrush** (social sentiment) were used to track how public perception and breaking news correlate with price volatility.
- **Regulatory & Institutional Portals** – Websites like **The Block Pro** or official government portals (e.g., SEC, EU MiCA updates) provided data on legal frameworks and institutional entry.

Cryptocurrency Landscape

- **Definition** – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
- **Market Diversity** – Thousands of cryptocurrencies exist, each serving unique roles:
    - **Store of Value (SoV)** – Bitcoin (BTC), characterized by digital scarcity.
    - **Utility Tokens** – Chainlink (LINK), Ether (ETH), used to pay for network services.

    - **Stablecoins** – USDT, USDC, pegged to fiat (USD) to provide price stability.
    - **Governance Tokens** – UNI, AAVE, granting holders voting rights over protocols.
- **Volatility** – Prices are highly sensitive to market sentiment, technological developments, and "macro" regulatory drivers.
- **Data Analysis Importance** – Integrating financial, social, and regulatory data is essential for a holistic understanding of asset performance and market health.

Data Collection Process

The program used the CoinGecko API to automate historical data fetching for Chainlink. The retrieved data was processed with Pandas to compute:

- **Daily Returns** – Percentage change between consecutive days to measure volatility.
- **Moving Averages (MA7, MA30)** – Short-term and long-term trend indicators.
- **Cumulative Returns** – Growth factor over time.
- **Rolling Volatility** – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using Matplotlib, a six-panel dashboard was created to display:

- **Price Trend** – Daily price movement over the selected period.
- **Price vs Volume** – Correlation between trading activity and price changes.
- **Daily Returns Distribution** – Statistical spread of daily gains/losses.
- **Moving Averages** – Technical analysis trend lines for price movement.
- **Cumulative Returns** – Overall growth performance.
- **Rolling Volatility** – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered advancements in blockchain technology, including:

- **Consensus Mechanisms** – The shift toward Proof of Stake (PoS) for sustainability and Proof of History (PoH) for faster processing.
- **Scalability Solutions** – Implementation of Layer-2 Rollups (Optimistic and ZK-Rollups) and Sharding to solve the "Blockchain Trilemma."
- **Interoperability Protocols** – Tools like Chainlink CCIP and Polkadot's Relay Chain enabling seamless cross-chain data and value exchange.
- **Privacy Features** – Zero-Knowledge Proofs (ZKP) allowing verifiable transactions without revealing sensitive underlying data.
- **Technical Roadmaps** – Ethereum's focus on "The Surge" (100k+ TPS) and Solana's "Firedancer" client for institutional-grade reliability.

Market Trends & Adoption

- **Merchant Acceptance** – Significant surge in interest with major processors like Stripe and Shopify integrating stablecoins for global settlement.
- **Wallet Growth** – Increasing downloads of non-custodial wallets (MetaMask, Phantom), indicating a shift toward direct on-chain interaction.
- **Network Activity** – Analysis of transaction volumes vs. active addresses to distinguish between "whale" activity and organic retail growth.
- **Integration with Traditional Finance** – The rise of Real-World Asset (RWA) Tokenization, bringing traditional assets like bonds and real estate on-chain.
- **Regulatory Developments** – The implementation of frameworks like the EU's MiCA and U.S. FIT21 Act, providing a clear licensing roadmap for the industry

Steps:

1. **Install Required Libraries**
   Install Python libraries such as `requests`, `pandas`, and `matplotlib` using pip to enable data fetching, processing, and visualization
2. **Obtain API Key**
   Register on the CryptoCompare website and generate an API key to access cryptocurrency market data.
3. **Set Configuration Parameters**
   Define the required parameters in the code:
   - Cryptocurrency symbol (e.g., LINK)
   - Comparison currency (e.g., USD)
   - Number of days for historical data
4. **Fetch Cryptocurrency Data**
   Use the API to retrieve historical market data including price and trading volume.
5. **Create DataFrame**
   Convert the fetched data into a Pandas DataFrame and format timestamps into readable date-time values.
6. **Process the Data**
   Perform calculations to analyze the data:
   - Daily Returns
   - Moving Averages (7-day and 30-day)
   - Cumulative Returns
   - Rolling Volatility
7. **Generate Visualization Dashboard**
   Use Matplotlib to create multiple graphs:
   - Price Trend
   - Price vs Volume
   - Returns Distribution
   - Moving Averages
   - Cumulative Returns
   - Volatility
8. **Execute the Program**
   Run the Python script to generate the analysis and visual outputs.
9. **Analyze the Results**
   Interpret the graphs to understand price trends, market behavior, and risk levels.
10. **Draw Conclusion**
    Summarize insights about cryptocurrency performance and the importance of data analysis in understanding the crypto market.

CODE:
Utilizing Api for analysis and visualization :

Code:

```
import requests
import matplotlib.pyplot as plt import
pandas as pd
import datetime

# ---------- CONFIGURATION ---------
# Replace 'YOUR_API_KEY' with the key you got from CryptoCompare
API_KEY = 'YOUR_API_KEY'
SYMBOL = 'LINK'
COMPARISON_SYMBOL =
'USD'
LIMIT = 180    # Number of days

# ---------- Fetch Data Function ----------
def fetch_cryptocompare_data(symbol, comparison_symbol, limit, api_key): url =
    f"https://min-api.cryptocompare.com/data/v2/histoday"
    params = {
        "fsym": symbol,
        "tsym": comparison_symbol,
        "limit": limit,
        "api_key": api_key
    }

    response = requests.get(url, params=params) data =
    response.json()

    if data['Response'] == 'Success':
        # CryptoCompare returns OHLCV data in 'Data' -> 'Data' return
        data['Data']['Data']

    else:
        print(f"Error fetching data: {data.get('Message')}") return None

# ---------- Dashboard Function ----------
def cryptocompare_dashboard(symbol, comparison_symbol, limit, api_key): raw_data =
```

fetch_cryptocompare_data(symbol, comparison_symbol, limit, api_key)
if not raw_data: return

 # Create DataFrame
df = pd.DataFrame(raw_data)

 # Convert 'time' (unix timestamp) to datetime df['timestamp'] = pd.to_datetime(df['time'], unit='s') df.set_index('timestamp', inplace=True)

 # Standardize column names to match your previous logic #
'close' is used as the daily price
df['price'] = df['close']
df['volume'] = df['volumeto'] # Total volume in USD (volumeto)

 # Calculate metrics
df['returns'] = df['price'].pct_change() df['MA7'] = df['price'].rolling(window=7).mean()
df['MA30'] = df['price'].rolling(window=30).mean() df['cumulative'] = (1 + df['returns']).cumprod() df['volatility'] = df['returns'].rolling(window=7).std() * 100

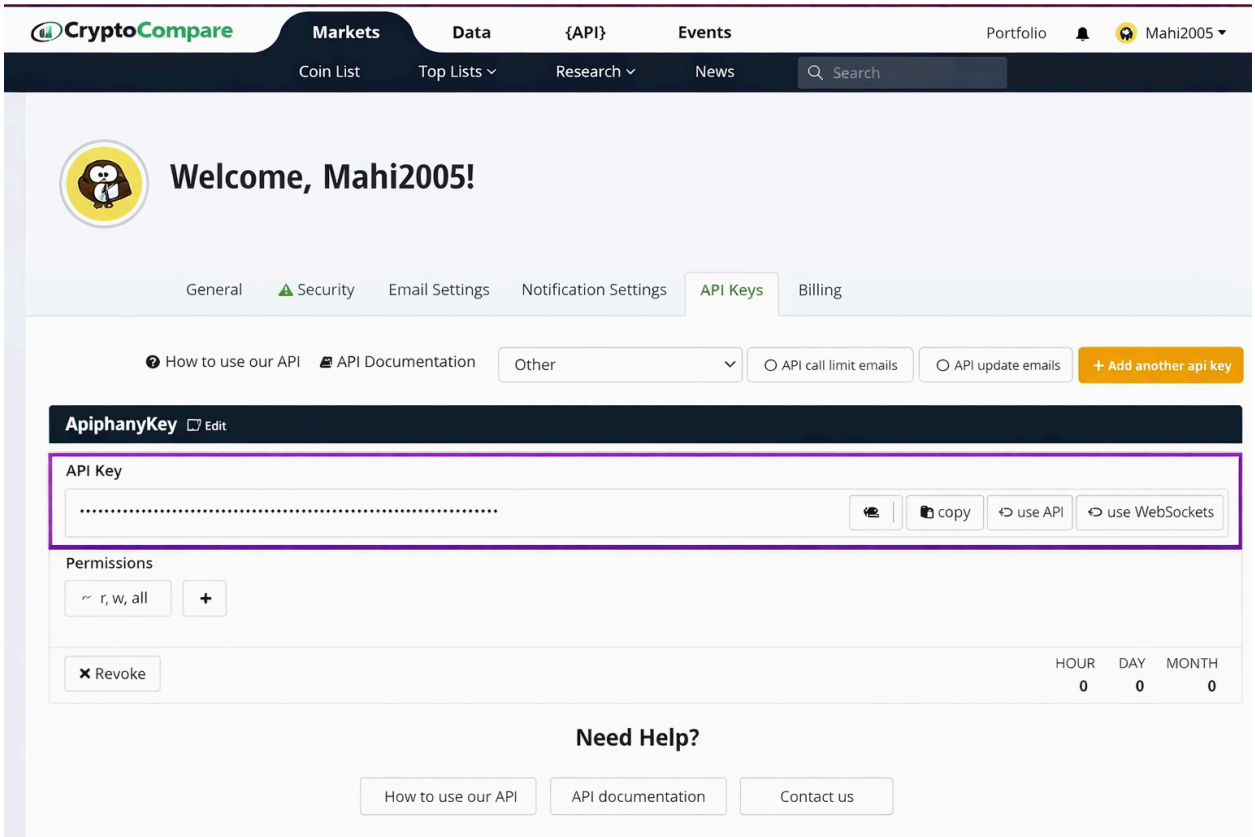
---------- **Plotting** ----------
fig, axes = plt.subplots(3, 2, figsize=(16, 12)) fig.suptitle(f'CryptoCompare: {symbol} Analysis Dashboard', fontsize=16, fontweight='bold')

 # 1. Price Trend
axes[0, 0].plot(df.index, df['price'], color='tab:blue', linewidth=2) axes[0, 0].set_title('Historical Price Trend (USD)')
axes[0, 0].grid(True, linestyle='--', alpha=0.7)

 # 2. Price vs Volume

```
axes[0, 1].plot(df.index, df['price'], color='tab:blue', label='Price')
axes[0, 1].bar(df.index, df['volume'], color='gray', alpha=0.3, label='Volume (USD)')
axes[0, 1].set_title('Price vs Trading Volume') axes[0,
1].legend()

# 3. Daily Returns Distribution
axes[1, 0].hist(df['returns'].dropna(), bins=40, color='tab:purple', edgecolor='white')
axes[1, 0].set_title('Daily Returns Distribution')

# 4. Moving Averages
axes[1, 1].plot(df.index, df['price'], color='lightgray', alpha=0.5, label='Price')
axes[1, 1].plot(df.index, df['MA7'], color='tab:orange', label='7-Day MA')
axes[1, 1].plot(df.index, df['MA30'], color='tab:red', label='30-Day MA')
axes[1, 1].set_title('Trend Analysis (Moving Averages)') axes[1, 1].legend()

# 5. Cumulative Returns
axes[2, 0].fill_between(df.index, df['cumulative'], color='tab:green', alpha=0.2)
axes[2, 0].plot(df.index, df['cumulative'], color='tab:green') axes[2,
0].set_title('Cumulative Growth')

# 6. Volatility
axes[2, 1].plot(df.index, df['volatility'], color='tab:red') axes[2,
1].set_title('7-Day Rolling Volatility (%)')

plt.tight_layout(rect=[0, 0.03, 1, 0.95]) plt.show()
```

OUTPUT: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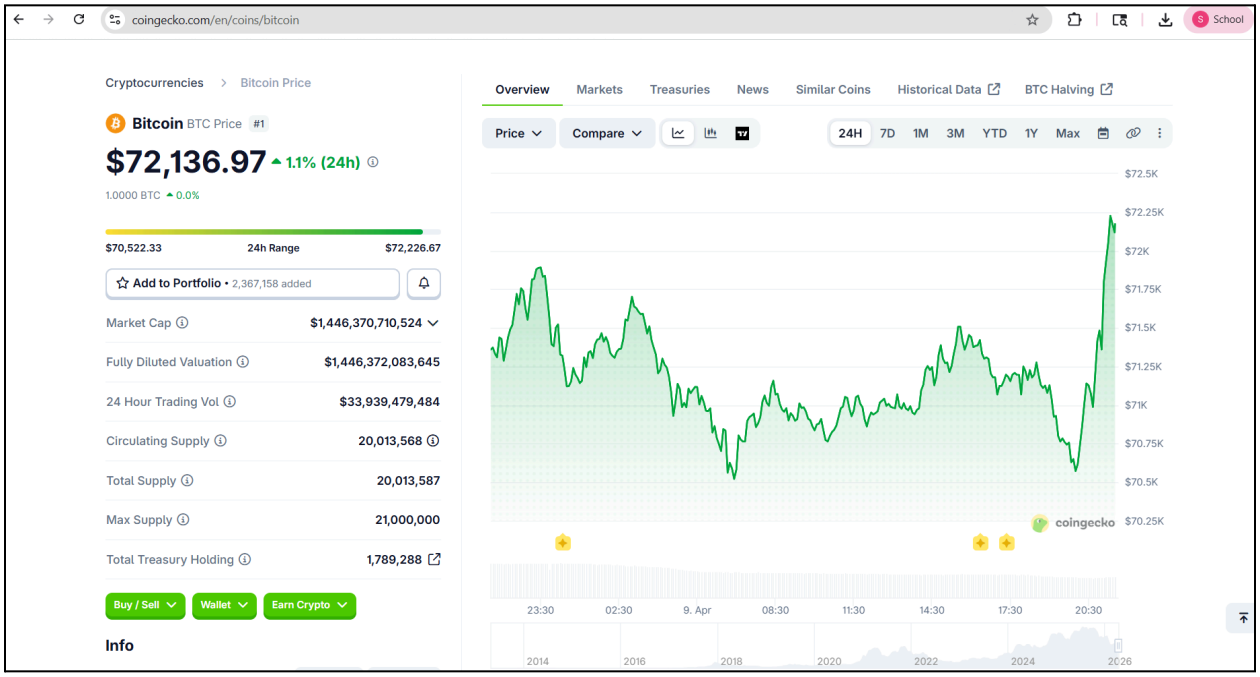
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Cryptocurrencies
Bitcoin Price
Bitcoin BTC Price #1
$72,136.97
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$70,522.33
24h Range
$72,226.67
Add to Portfolio • 2,367,158 added
Market Cap $1,446,370,710,524
Fully Diluted Valuation $1,446,372,083,645
24 Hour Trading Vol $33,939,479,484
Circulating Supply 20,013,568
Total Supply 20,013,587
Max Supply 21,000,000
Total Treasury Holding 1,789,288
Buy / Sell
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Info
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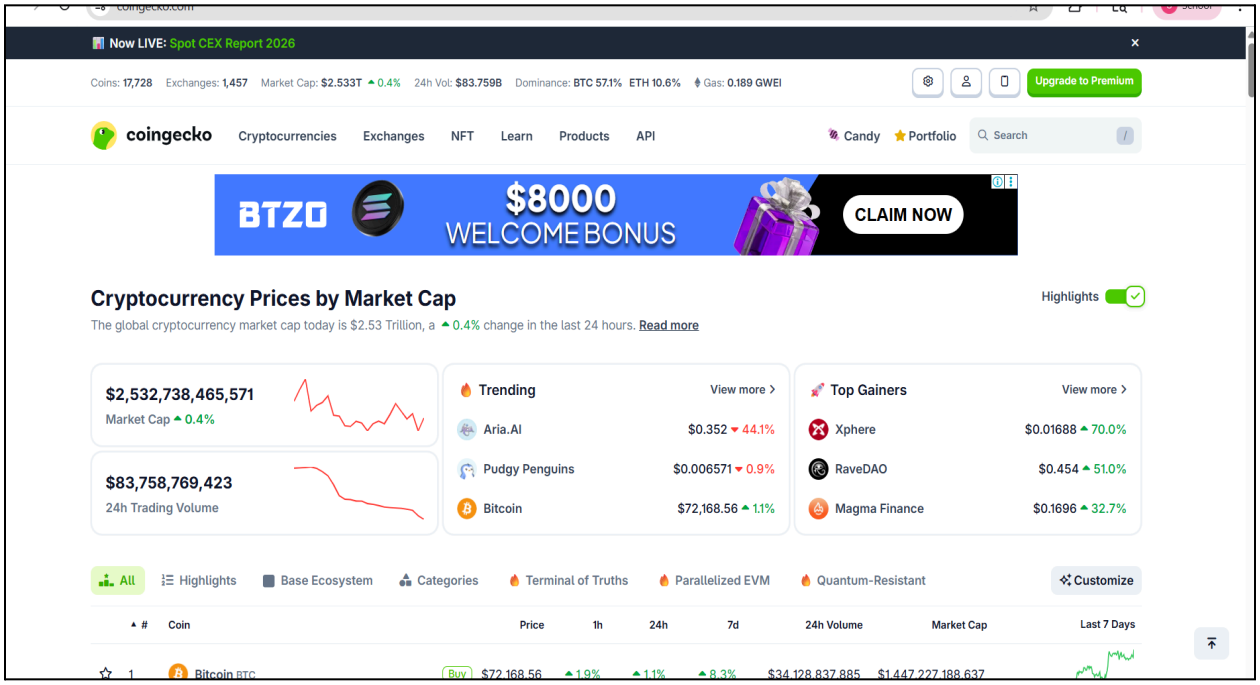
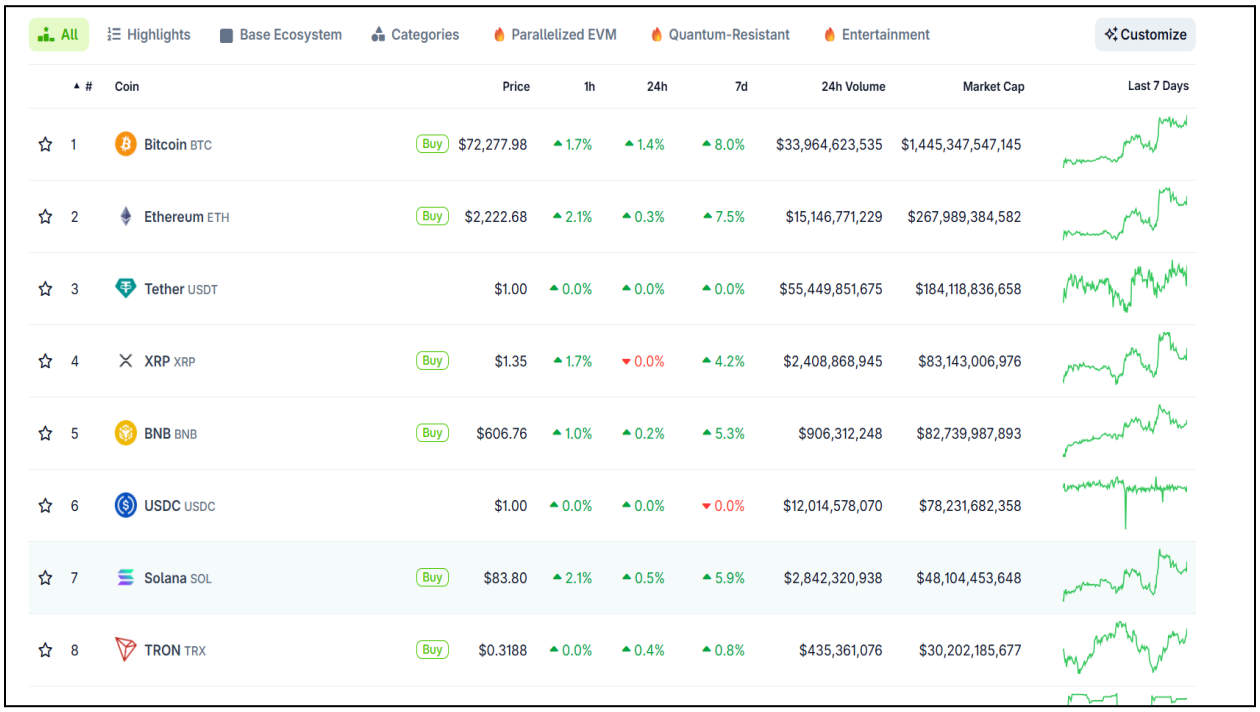
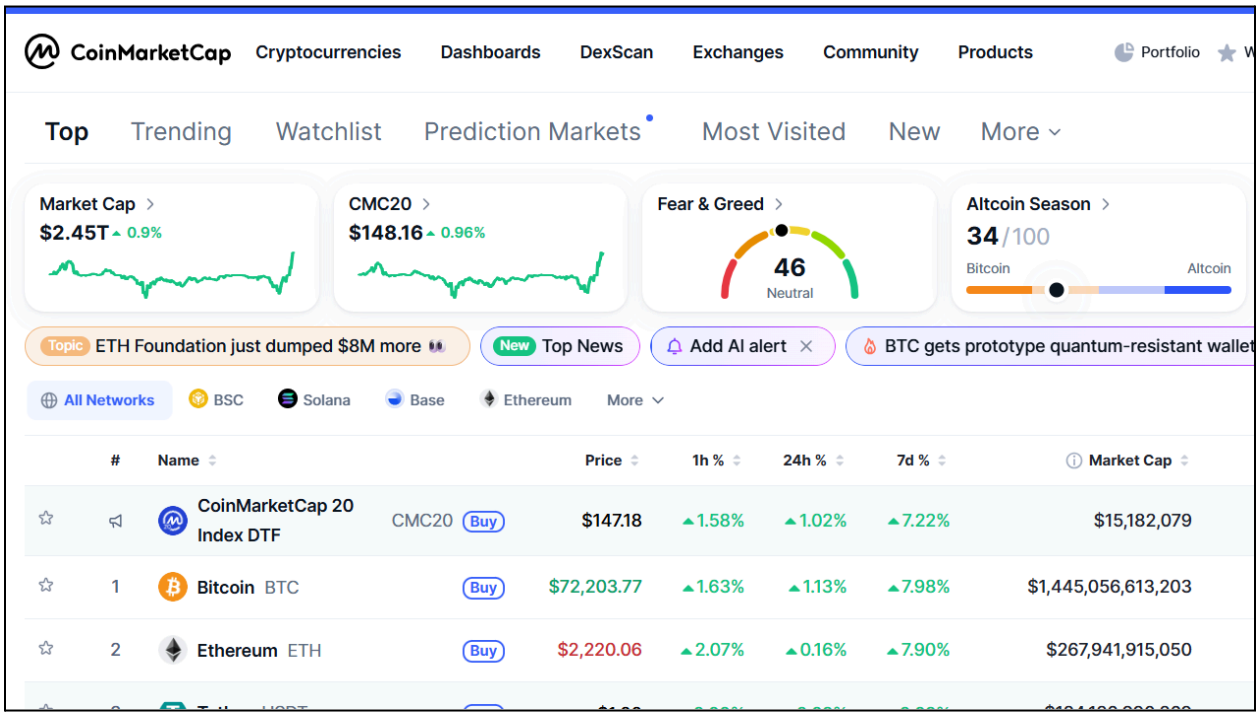




All | Highlights | Base Ecosystem | Categories | Parallelized EVM | Quantum-Resistant | Entertainment | Customize

#	Coin		Price	1h	24h	7d	24h Volume	Market Cap	Last 7 Days
1	Bitcoin BTC	Buy	$72,277.98	1.7%	1.4%	8.0%	$33,964,623,535	$1,445,347,547,145	
2	Ethereum ETH	Buy	$2,222.68	2.1%	0.3%	7.5%	$15,146,771,229	$267,989,384,582	
3	Tether USDT		$1.00	0.0%	0.0%	0.0%	$55,449,851,675	$184,118,836,658	
4	XRP XRP	Buy	$1.35	1.7%	0.0%	4.2%	$2,408,868,945	$83,143,006,976	
5	BNB BNB	Buy	$606.76	1.0%	0.2%	5.3%	$906,312,248	$82,739,987,893	
6	USDC USDC		$1.00	0.0%	0.0%	0.0%	$12,014,578,070	$78,231,682,358	
7	Solana SOL	Buy	$83.80	2.1%	0.5%	5.9%	$2,842,320,938	$48,104,453,648	
8	TRON TRX	Buy	$0.3188	0.0%	0.4%	0.8%	$435,361,076	$30,202,185,677	


CoinMarketCap
Cryptocurrencies
Dashboards
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Market Cap
$2.45T 0.9%
CMC20
$148.16 0.96%
Fear & Greed
46
Neutral
Altcoin Season
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All Networks
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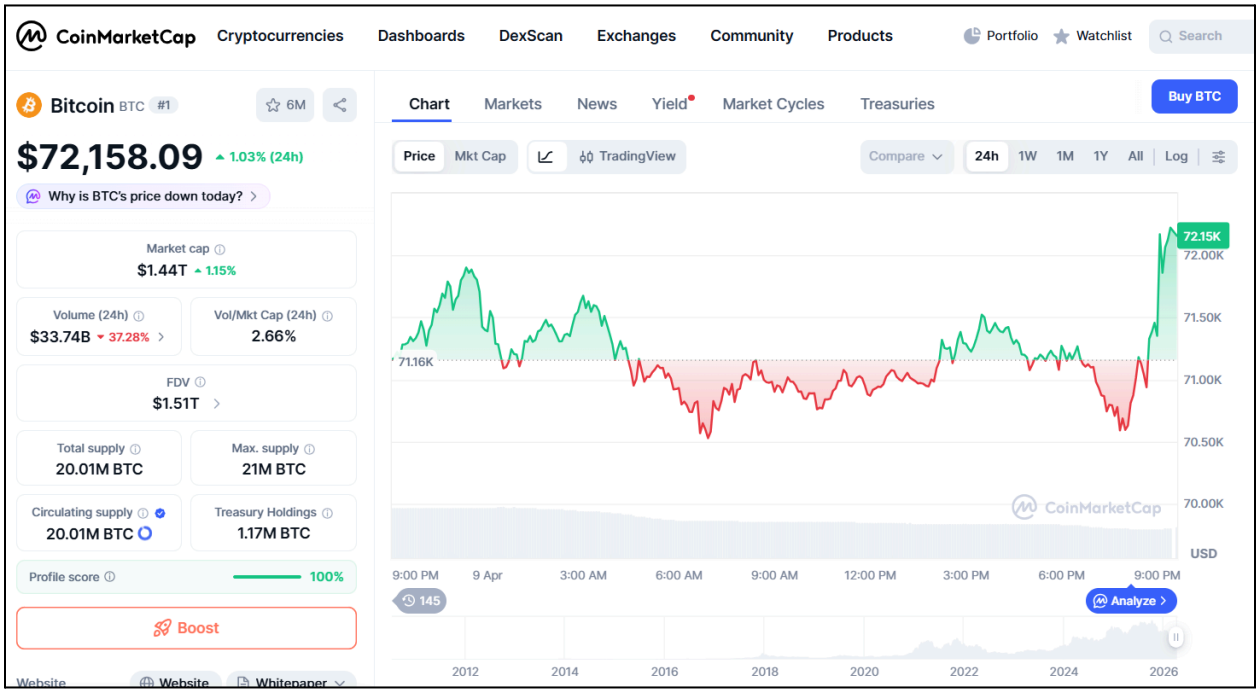
Name Price 1h % 24h % 7d % Market Cap
CoinMarketCap 20 Index DTF CMC20 Buy $147.18 1.58% 1.02% 7.22% $15,182,079
1 Bitcoin BTC Buy $72,203.77 1.63% 1.13% 7.98% $1,445,056,613,203
2 Ethereum ETH Buy $2,220.06 2.07% 0.16% 7.90% $267,941,915,050


CoinMarketCap
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Bitcoin BTC #1
6M
$72,158.09 1.03% (24h)
Why is BTC's price down today?
Market cap
$1.44T 1.15%
Volume (24h)
$33.74B 37.28%
Vol/Mkt Cap (24h)
2.66%
FDV
$1.51T
Total supply
20.01M BTC
Max. supply
21M BTC
Circulating supply
20.01M BTC
Treasury Holdings
1.17M BTC
Profile score
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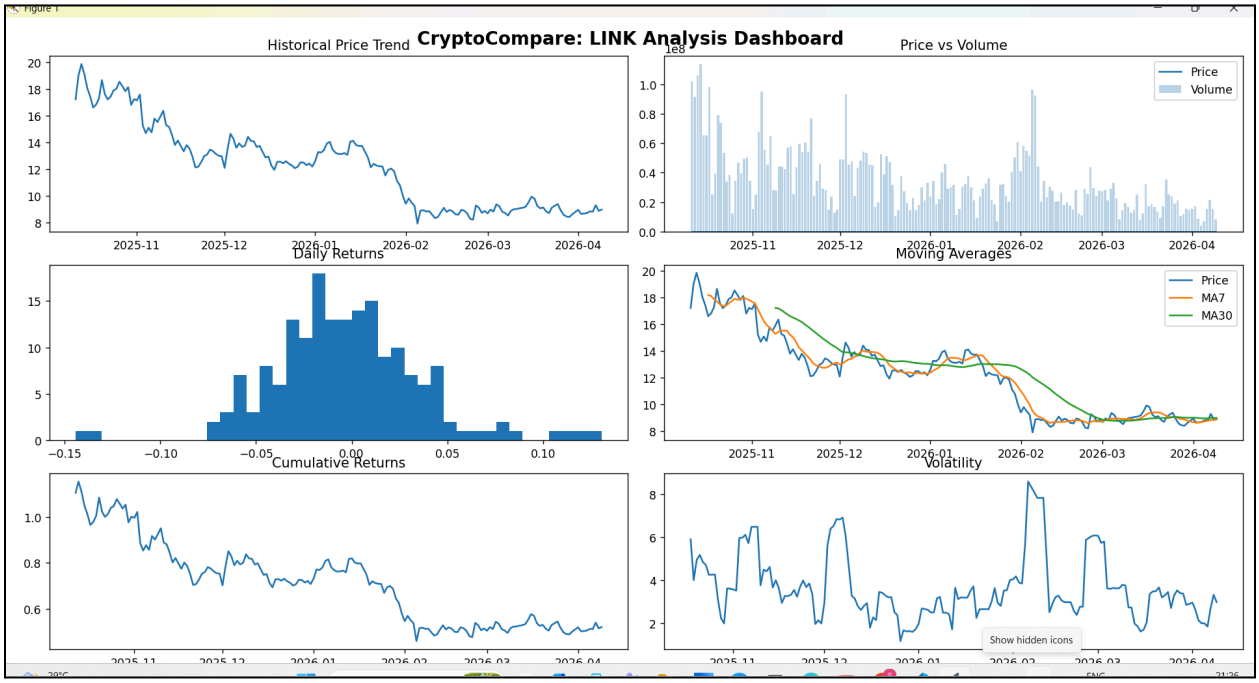

Conclusion:-

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.